Statistical Analysis of HateCrimes in USA

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What is a Hate Crime?

A crime motivated by bias or prejudice against a person's:

- Race
- Religion
- Gender / Sexual orientation
- Disability, ethnicity, or other protected characteristic

It can involve assault, threats, or property damage.

Hate Crimes in the USA:

Nature:

Physical, verbal, property, or online offenses

Estimated Victims:

In 2024, 14,243 victims were affected in 11,679 reported incidents (FBI, UCR Program, 2025).

Trends(Reported Hate Crime Totals, 2023 and 2024):

Category	2024	2023
Incidents	11,679	11,862
Offenses	"13,683"	"13,829"
Victims	"14,243"	"14,416"
Known Offenders	"10,096"	"9,739"

Source:

FBI Hate Crime Data

Size:

296 victims

Purpose:

Analyze patterns and relationships between features like:

- Place
- Offense type
- Bias type

Goal:

Understand distributions and correlations in hate crimes.

- Distribution of victims by bias type.
- Relationship between age and type of offense.
- Probability patterns of hate crime types.

Features to Analyze:

- Age
- Offense Type
- Bias
- Location

Relationships/Questions:

- Does age affect likelihood of being targeted?
- Which bias type is most common?
- Relationship between offense type and bias.

Data Description:

The dataset contains recorded hate crime incidents with information on:

- Bias motivation
- Offense type
- Location of incident
- Number of victims and offenders (under and over 18)
- Temporal attributes such as day of the week

Both categorical and numerical variables are included, allowing descriptive, probabilistic, and inferential statistical analysis.

	Month	Incident Number	Day of Week	Number of Victims under 18	Number of Victims over 18	Number of Offenders under 18	Number of Offenders over 18	Race/Ethnicity of Offenders	Offense(s)	Offense Location	Bias
0	Jan	2017-241137	Sun	0	1	0.0	1.0	White/Not Hispanic	Aggravated Assault	Park/Playground	Race/Ethnicity-Based
1	Feb	2017-580344	Wed	0	1	0.0	1.0	Black or African American/Not Hispanic	Aggravated Assault	Highway/Road/Alley/Street/Sidewalk	Race/Ethnicity-Based
2	Mar	2017-800291	Tue	0	0	0.0	0.0	Unknown	Destruction	Highway/Road/Alley/Street/Sidewalk	Religion-Based
3	Apr	2017-1021534	Wed	0	0	0.0	0.0	White/Unknown	Simple Assault	Air/Bus/Train Terminal	Religion-Based
4	May	2017-1351550	Mon	1	0	1.0	2.0	White/Not Hispanic	Simple Assault	Residence/Home	Sexual Orientation-Based

Descriptive Statistics:

Frequency Distributions:

Frequency and percentage analyses were conducted for:

- Bias motivation
- Offense type
- Location

Bias motivation:

Which bias type is most common?

📊 **Distribution of Hate Crimes by Bias Group**

	Bias Group	Count	Percentage
0	Race/Ethnicity-Based	111	37.63%
1	Other Bias Category	67	22.71%
2	Sexual Orientation-Based	59	20.00%
3	Religion-Based	41	13.90%
4	Gender/Gender Identity-Based	17	5.76%

Conclusion:

Race/Ethnicity-Based bias accounts for the majority of hate crimes (37.63%), followed by Other Bias (22.71%) and Sexual Orientation-Based bias (20%).

Offense type:

Which offense type is most common?

Top 7 Offense Types

	Offense Type	Count	Percentage
0	Criminal Mischief	56	18.98%
1	Assault	53	17.97%
2	Assault with Injury	28	9.49%
3	Terroristic Threat	27	9.15%
4	Aggravated Assault	26	8.81%
5	Harassment	25	8.47%
6	Assault by Threat	13	4.41%
7	Other	8	2.71%

Conclusion:

Criminal Mischief (18.98%) and Assault (17.97%) are the most frequent offense types, together accounting for over one-third of incidents, while less common offenses like Assault by Threat (4.41%) and Other (2.71%) occur relatively rarely.

Location:

Does the location of the incident influence the likelihood of hate crimes occurring?

Top 7 Locations of Hate Crimes (Other: count=2)

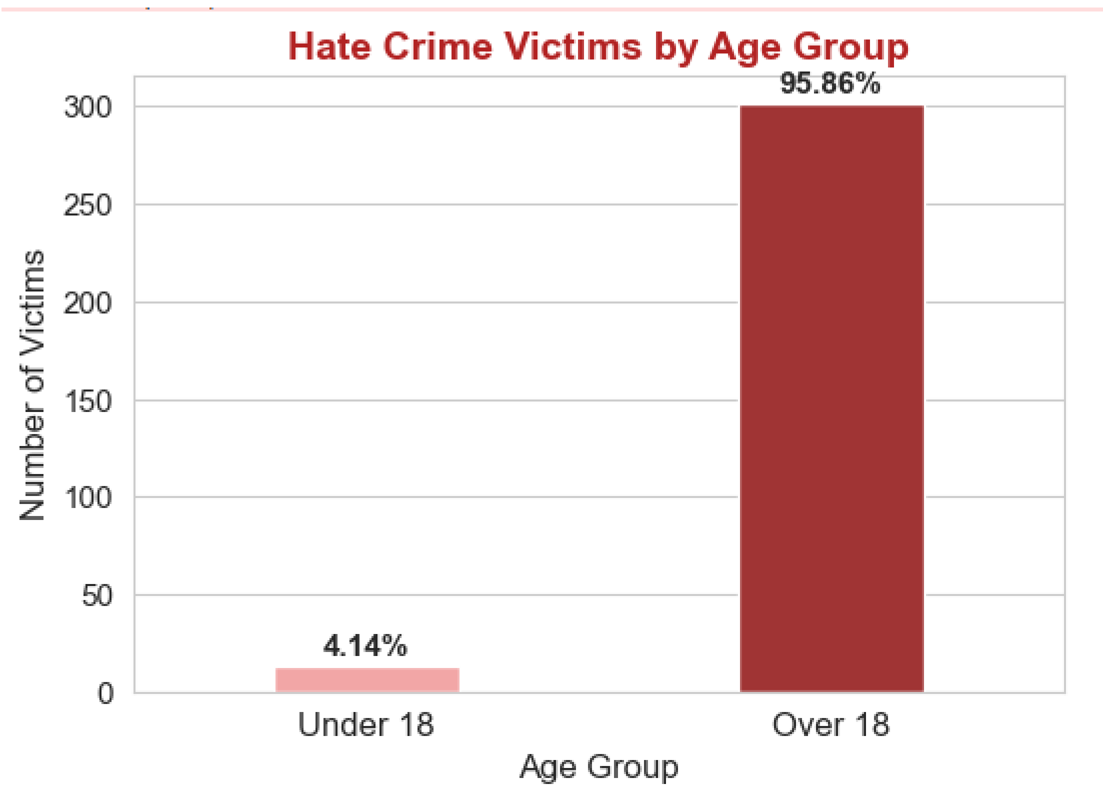
	Location	Count	Percentage
0	residence/home	75	25.17%
1	street/road	65	21.81%
2	parking/garage	24	8.05%
3	bar/nightclub	19	6.38%
4	church/synagogue/temple/mosque	17	5.70%
5	Other	14	4.70%
6	commercial/office building	11	3.69%
7	park/playground	8	2.68%

Conclusion:

The majority of hate crime incidents take place at residences or homes (25.17%) and on streets or roads (21.81%), whereas parks, playgrounds (2.68%), and commercial or office buildings (3.69%) are among the least affected locations.

Victim Age Analysis:

Does age affect likelihood of being targeted?



Conclusion:

Yes, age appears to affect the likelihood of being targeted, with adults over 18 being far more frequently victimized (95.86%) compared to minors under 18 (4.14%), indicating that hate crimes predominantly affect the adult population.

Probability Analysis:

Conditional probability analysis was conducted to examine the likelihood of specific offense types given a particular bias motivation.

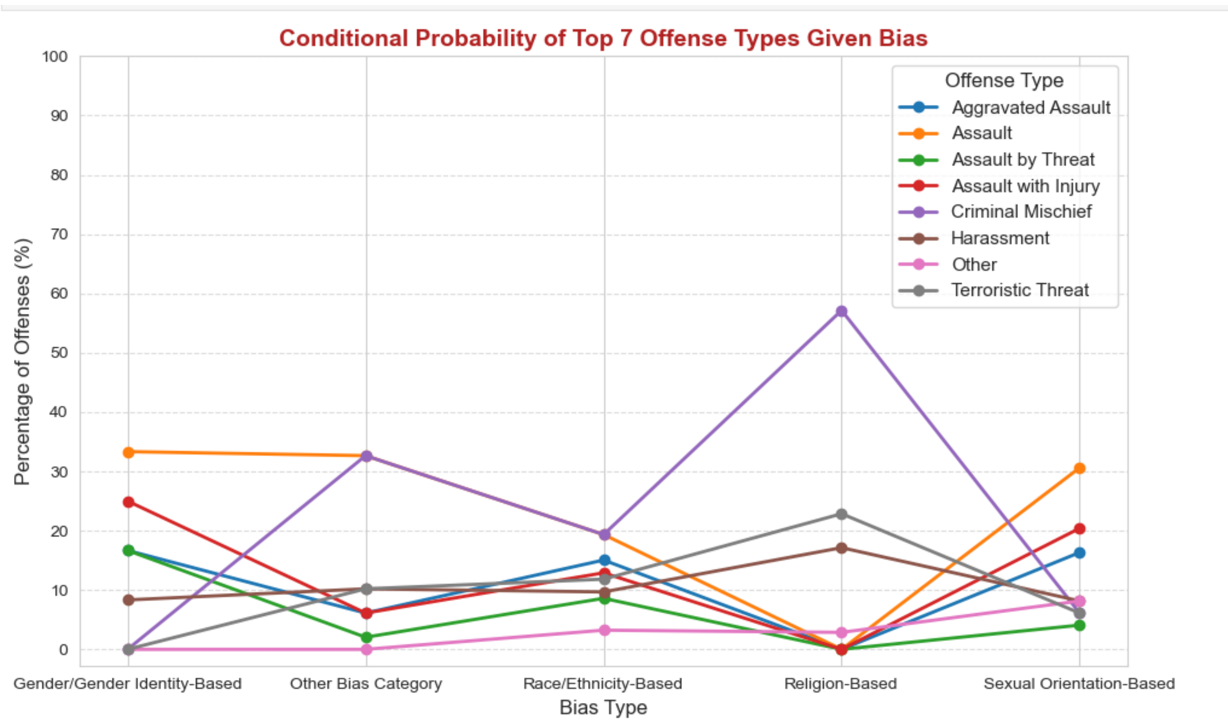
Conditional Probability of Top 7 Offenses Given Bias

Offense Top	Bias Type	Aggravated Assault	Assault	Assault by Threat	Assault with Injury	Criminal Mischief	Harassment	Terroristic Threat
0	Gender/Gender Identity-Based	16.67%	33.33%	16.67%	25.00%	0.00%	8.33%	0.00%
1	Other Bias Category	6.12%	32.65%	2.04%	6.12%	32.65%	10.20%	10.20%
2	Race/Ethnicity-Based	15.56%	20.00%	8.89%	13.33%	20.00%	10.00%	12.22%
3	Religion-Based	0.00%	0.00%	0.00%	0.00%	58.82%	17.65%	23.53%
4	Sexual Orientation-Based	17.78%	33.33%	4.44%	22.22%	6.67%	8.89%	6.67%

Conclusion:

The data indicates a clear relationship between offense type and bias, with certain offenses occurring more frequently under specific biases—for example, Religion-Based incidents are largely concentrated in one category, while Gender/Gender Identity and Sexual Orientation-Based offenses are more evenly distributed across multiple offense types.

Graphical Representation:



Advanced Statistical Analysis:

Z-Score:

Z-scores were calculated to standardize victim and offender counts and identify extreme incidents (|Z| > 3).

$$Z = \frac{X - \mu}{\sigma}$$

Parameters for 'Number of Victims over 18':

- Mean ((\mu)) = 0.39
- Standard Deviation ((\sigma)) = 0.53
- Example: Incident 80 with 3 victims →

$$Z = \frac{3 - 0.39}{0.53} \approx 4.92$$

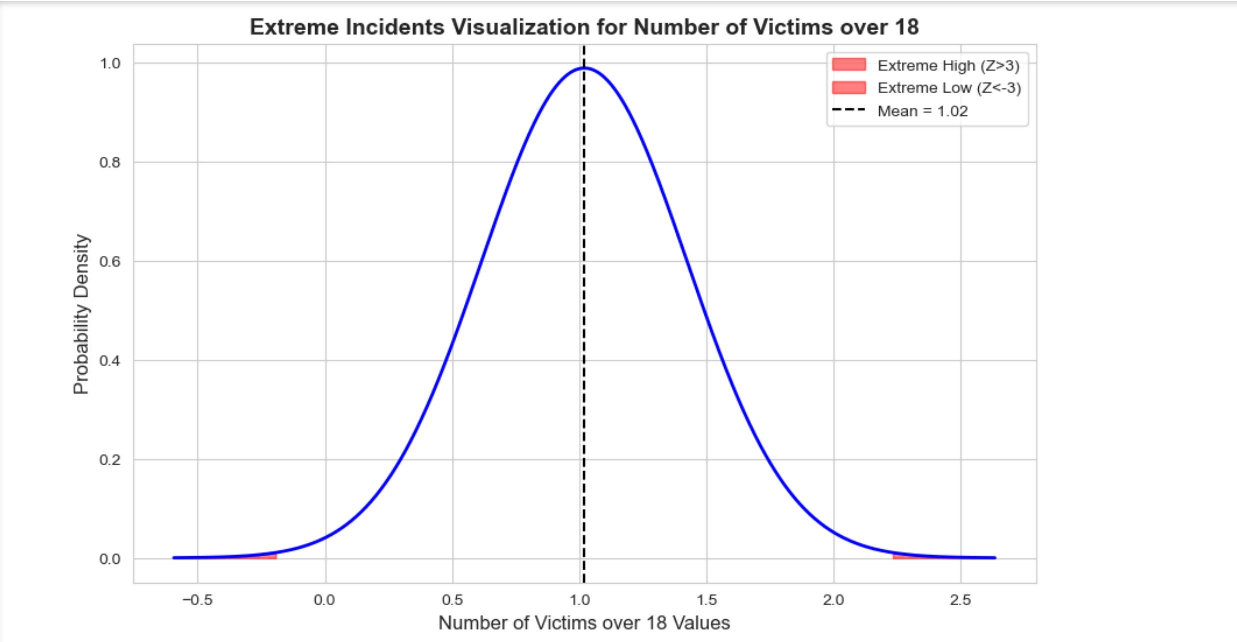
Extreme Incidents (|Z| > 3):

	Incident ID	Number of Victims over 18	Z-Score
0	80	3	4.924528
1	207	3	4.924528
2	221	3	4.924528
3	294	3	4.924528

Conclusion:

The calculated Z-scores for incidents with 3 victims over 18 (Z≈4.91Z \approx 4.91Z≈4.91) are well above the threshold of |Z|>3|Z| > 3|Z|>3, indicating that these incidents are **extreme outliers**. This means that having three adult victims is unusually high compared to the typical number of victims, and the probability of such an event occurring under normal circumstances is **very low**, highlighting these cases as **exceptionally severe hate crime incidents**.

Graphical Representation:



Confidence Intervals:

Confidence intervals for the mean number of victims under and over 18 were computed using a 95% confidence level.

$$\text{Confidence Interval: } \bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Parameters:

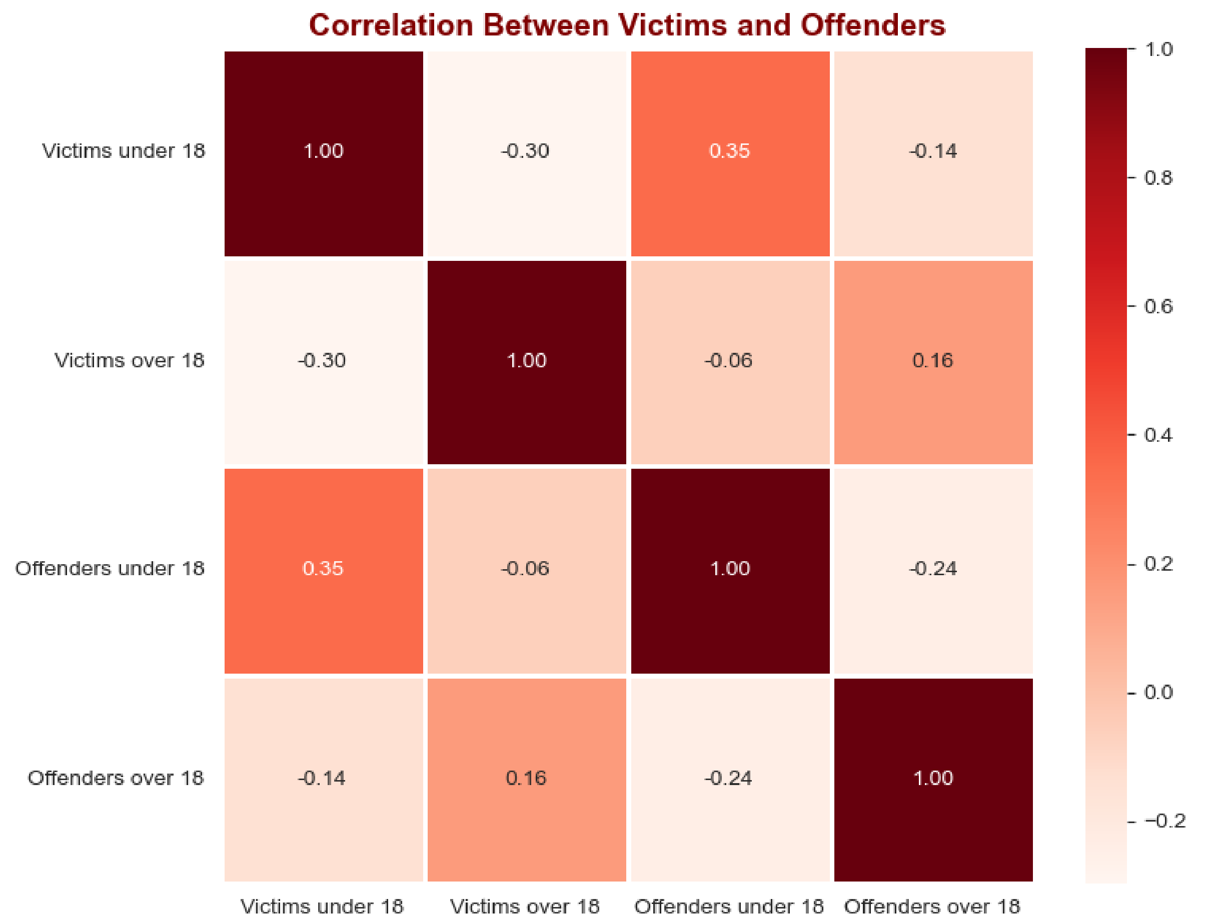
- Sample Mean ((\bar{X})) = 1.02
- Sample Standard Deviation (σ)= 0.40
- Sample Size (n) = 295
- Confidence Level = 95%
- Z-score ((Z_{\alpha/2})) = 1.96
- Margin of Error = 0.05

Confidence Interval = (0.97, 1.07)

Conclusion:

Based on the 95% confidence interval of (0.97, 1.07) for the number of victims over 18, we can be reasonably confident that the true average number of adult victims per incident lies within this range, indicating that most incidents typically involve around 1 adult victim.

Correlation HeatMap:



Conclusion:

The heatmap shows mostly weak correlations, indicating that age alone does not strongly drive hate crime patterns.
There is a moderate positive relationship between victims under 18 and offenders under 18, suggesting youth-related incidents often involve peers.
Overall, hate crimes tend to occur within similar age groups, but other factors likely play a larger role than age.

Inferential Statistical Analysis

One-Sample t-Test Analysis (Gender-Based Hate Crimes)

Variable:
X = Number of victims per hate crime incident.

Reference Population:
All hate crime incidents in the dataset.

Population Mean:
μ_0 = Overall mean number of victims per incident.

Sample:
Gender-based incidents (Gender Identity + Sexual Orientation).
Sample size: $n = 76$ incidents.

Sample Mean:
$\bar{x}$_gender = 1.092 victims per incident.

Hypotheses:
H_0: μ_gender = μ_0
H_1: μ_gender ≠ μ_0

Test Statistic:
$t = (\bar{x}\_gender - \mu_0) / (s / \sqrt{n})$

Decision Rule:
Reject H_0 if p-value < 0.05.

The one-sample t-statistic is defined as:

$$t = \frac{\bar{X}_{\text{gender}} - \mu_0}{s_{\text{gender}} / \sqrt{n_{\text{gender}}}}.$$

the difference from the reference mean is evaluated directly:
s (standard deviation) = 0.3338

$$\bar{X}_{\text{gender}} - \mu_0 = 1.092 - 1.057 = 0.035.$$

this difference is small relative to the scale of the data

At significance level:

$$\alpha = 0.05$$

Gender-Based Hate Crimes (n=76):
Sample mean = 1.092
Reference μ_0 = 1.057
t-statistic = 0.916
p-value = 0.3628
Conclusion: FAIL to reject H_0 ($p > 0.05$)

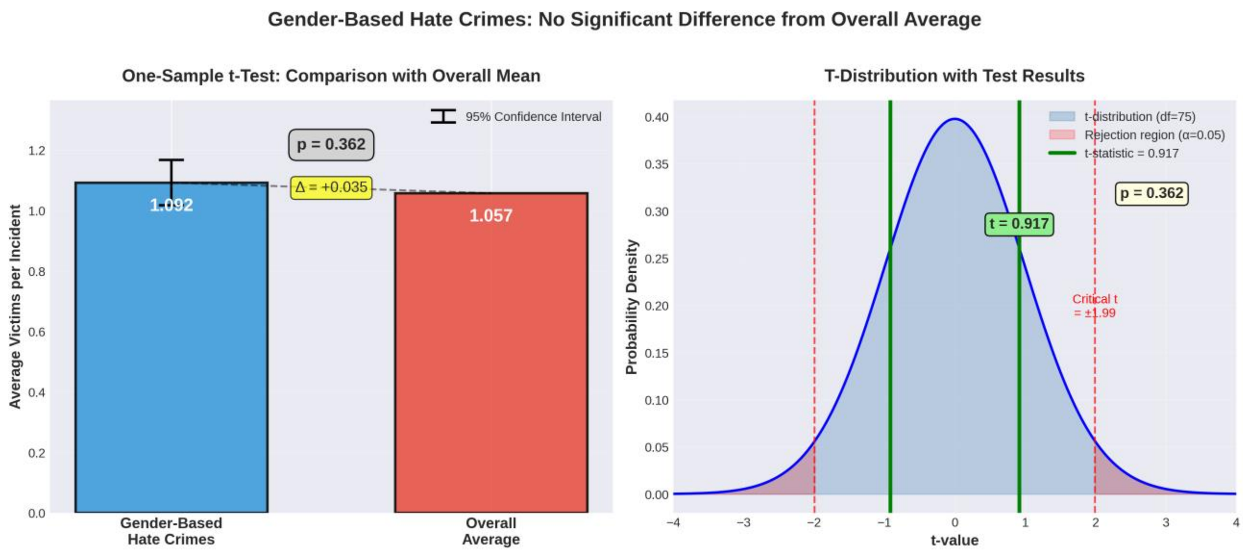
Fail to reject H_0

Results: Fails to reject Ho.

Interpretation:

There is insufficient statistical evidence to conclude that the mean number of victims per gender-based hate crime incident differs from the overall average number of victims per incident

Conclusion:
The one-sample t-test compares the average number of victims per gender-based hate crime incident with the overall average across all incidents. The result indicates *whether gender-based hate crimes involve a statistically different number of victims per incident compared to the general population of hate crimes.*


Gender-Based Hate Crimes: No Significant Difference from Overall Average
One-Sample t-Test: Comparison with Overall Mean
95% Confidence Interval
p = 0.362
Δ = +0.035
1.092
1.057
Average Victims per Incident
Gender-Based Hate Crimes
Overall Average
T-Distribution with Test Results
t-distribution (df=75)
Rejection region (α=0.05)
t-statistic = 0.917
p = 0.362
t = 0.917
Critical t = ±1.99
Probability Density
t-value

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Simple Linear Regression / Correlation Analysis

Average Victims per Incident vs Time

Variables Used:
Response Variable (Y):
Average number of victims per incident per month.

Predictor Variable (X):
Time index ($t = 1, 2, 3, \ldots$), representing consecutive months in the study period.

Purpose:
To examine whether there is a linear relationship between time and the average number of victims per incident, i.e., whether incidents tend to involve more or fewer victims as time progresses.

Model Specification:
$Y_t = \beta_0 + \beta_1 t + \varepsilon_t$

where:
β_0 = intercept (estimated average victims at the initial time),
β_1 = slope (average monthly change in victims per incident),
ε_t = random error term.

Hypotheses:
$H_0: \beta_1 = 0$
(No linear relationship between time and average victims per incident)

$H_1: \beta_1 \neq 0$
(A significant linear relationship exists)

Estimation Method:
Parameters β_0 and β_1 are estimated using the method of least squares.

Slope Estimator:
$\hat{\beta}_1 = \Sigma(t - \bar{t})(Y_t - \bar{Y}) / \Sigma(t - \bar{t})^2$

Intercept Estimator:
$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{t}$

Decision Rule:

Reject H_0 if the p-value associated with $\hat{\beta}_1$ is less than 0.05.

Interpretation of Results:
If $\hat{\beta}_1 > 0$ and statistically significant, the average number of victims per incident increases over time.
If $\hat{\beta}_1 < 0$ and statistically significant, the average number of victims per incident decreases over time.
If $\hat{\beta}_1$ is not statistically significant, no meaningful time trend is detected.

Correlation Interpretation:
The sign of $\hat{\beta}_1$ indicates the direction of correlation between time and average victims, while the magnitude reflects the strength of the linear relationship.

Coefficient of Determination (R^2):
R^2 represents the proportion of variability in average victims per incident that is explained by time.

Correlation Analysis Calculations:

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For each month t:

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$$\text{AvgVictims}_t = \frac{\text{Total Victims in month } t}{\text{Number of incidents in month } t}$$

n = is the total number of months in the dataset

$$\{(t, \text{AvgVictims}_t) \mid t = 1, 2, \ldots, n\}.$$

Siimple linear regression

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

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$$x_i = Monthnumber(1, 2, 3, \ldots upto 106)$$

$$y_i = Averagevictimsforthatmonth\bar{x} = Averagemonthnumber = 53.5$$

$$\bar{y} = Overallaveragevictimspermonth$$

$$n = Numberofmonths = 106$$

$$\begin{aligned} \sum (x_i - \bar{x})^2 &= 99225 \\ \sum (y_i - \bar{y})^2 &= 68.024 \\ \sum (x_i - \bar{x})(y_i - \bar{y}) &= -156.43 \end{aligned}$$

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Plugging into formula

$$r = \frac{-156.43}{\sqrt{99225 \times 68.024}} = \frac{-156.43}{\sqrt{6,748,492}} = \frac{-156.43}{2597.79} = -0.0602$$

Interpretation:

Correlation coefficient(r) = -0.0602

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What -0.0602 means

- ***Direction:*** negative (slightly)
- ***Strength:*** weak (almost zero)
- ***In Simple:*** There's almost no relationship between time and average victims

$\|\mathbf{r}\|$ **value**	**Strength**
$0.00 - 0.19$	$Veryweak$
$0.20 - 0.39$	$Weak$
$0.40 - 0.59$	$Moderate$
$0.60 - 0.79$	$Strong$
$0.80 - 1.00$	$Verystrong$

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Answer:

Average victims per incident hasn't really changed from 2017 to 2025

2017 Average: 0.83 victims/incident
2025 Average: 1.00 victims/incident
Change: +0.17 over 9 years (tiny!)

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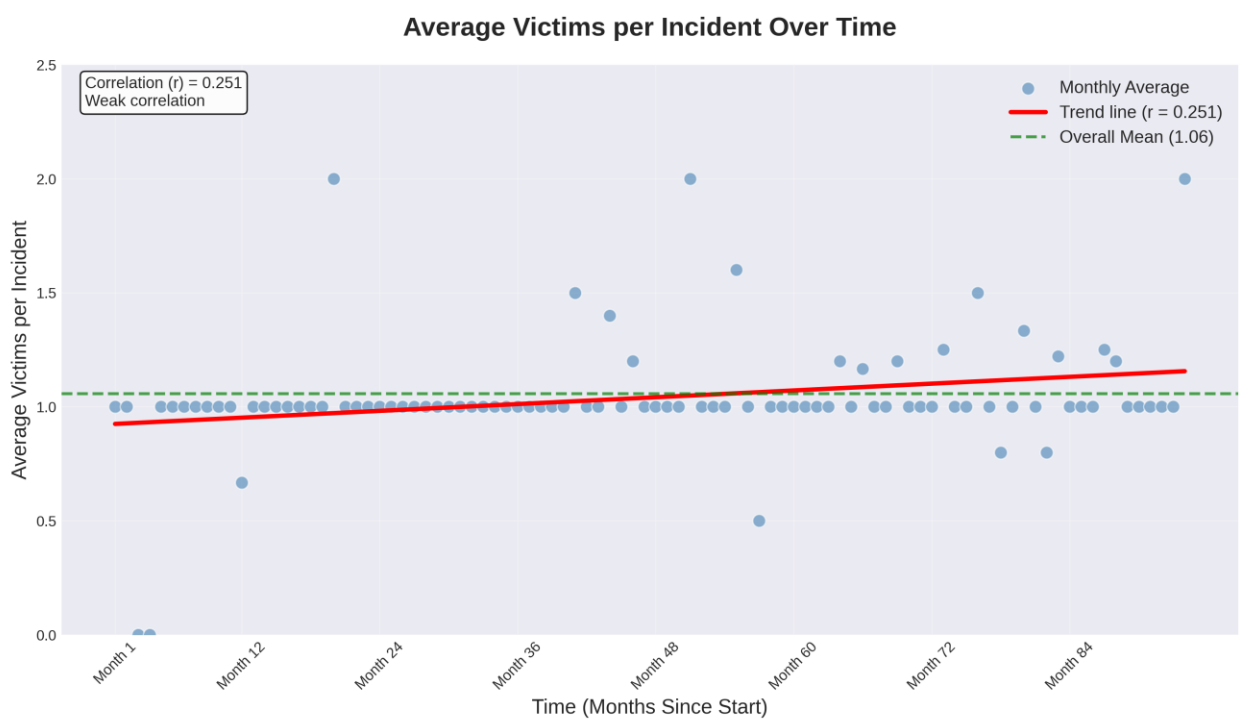

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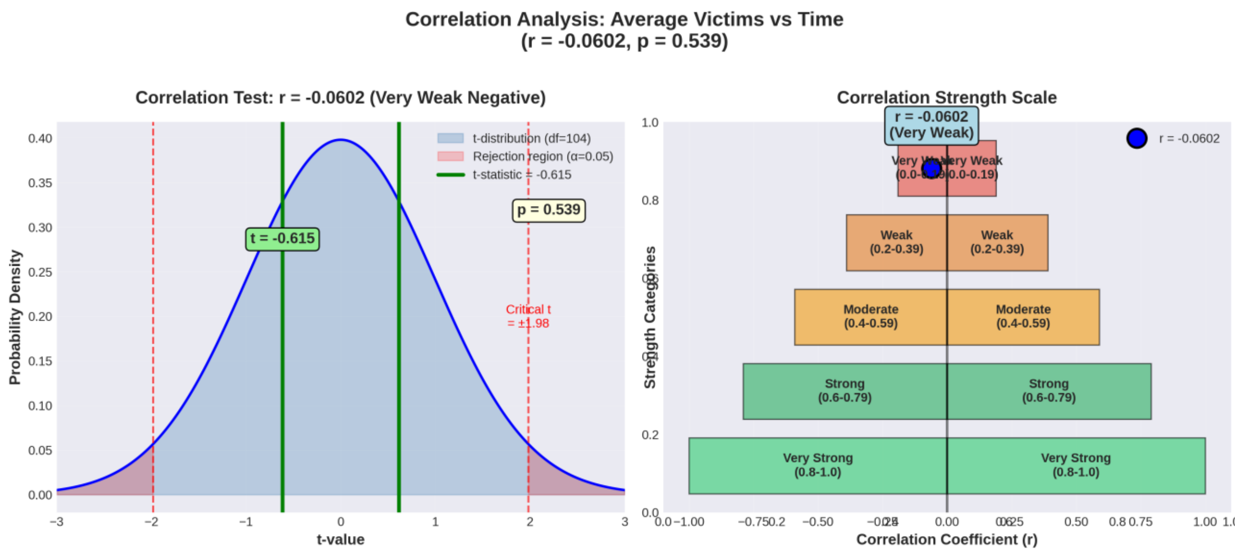

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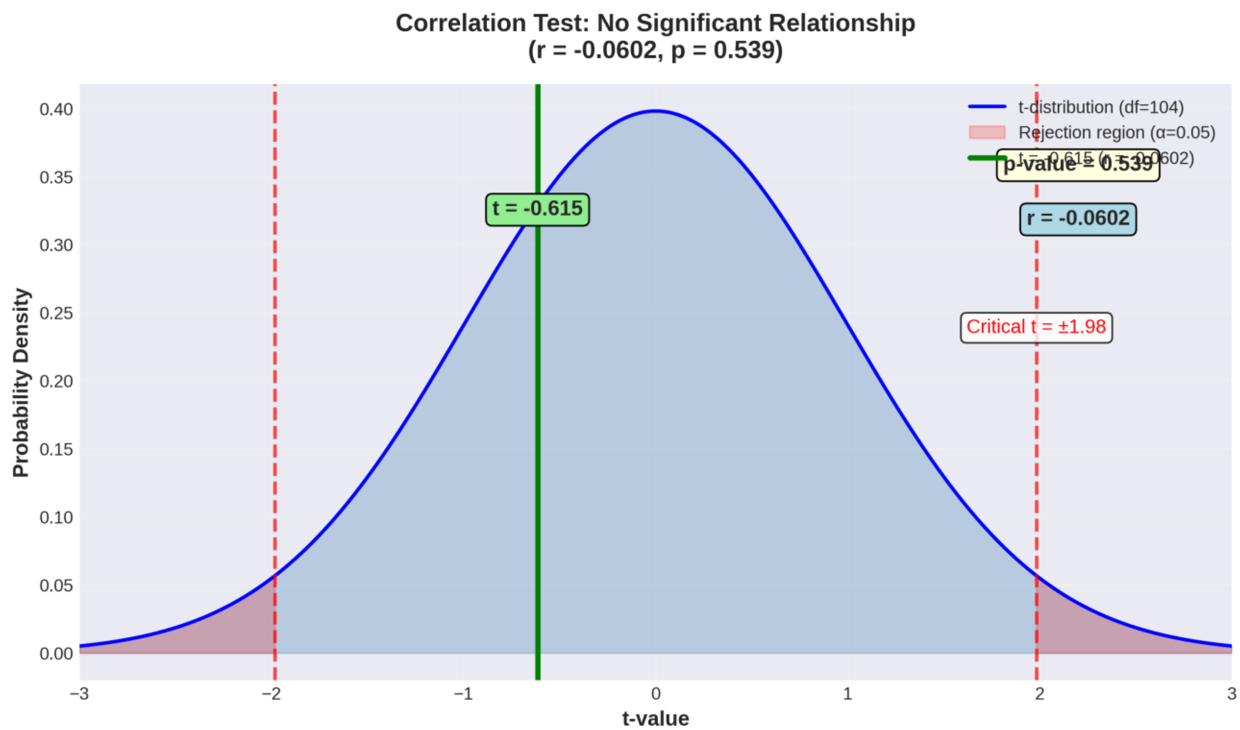

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Conclusion:

The results overall show that there is no much difference in average victim# counts over time ((when time passes)).The slope was not significant at the 5% level, indicating no clear upward or downward trend. Overall, incident severity remained relatively stable over time.